Case 38: Static live-fuel moisture sweep

1 Verification purpose

This case verifies live-fuel moisture damping and calculated live extinction while fuel loads remain fixed. The static shrub models SH2, SH6, and SH8 are intentionally used so changing live moisture cannot transfer herbaceous load into a dead class. It can expose live/dead index errors, percent conversion errors, incorrect live-extinction calculation, or unintended dynamic curing. The question is whether all static-fuel simulation conditions match independent I_R and rate of spread (ROS) predictions over 30 to 200%.

2 Mathematical formulation and numerical implementation

The live term uses the same cubic damping function $\eta_M(r_M)$ as dead fuel, but $r_M = M_L/M_{x,L}$ and $M_{x,L}$ is calculated from dead/live weighting and dead moisture rather than read directly. Live moisture also appears in $Q_{ig} = 250 + 1116M_f$. The formulation follows Andrews (2018), RMRS-GTR-371. For a static fuel, W_{1h}, W_{LH}, and W_{LW} are invariant throughout the sweep.

3 Derivation of expected behavior

The reference calculation first calculates the live extinction moisture for each frozen fuel row and fixed dead moisture. It then evaluates live damping, reaction intensity, heat sink, and R_0 at each moisture. Since no loads move, changes arise only from moisture physics. This controlled result is the baseline needed to interpret the additional load-transfer mechanism in CASE39.

Live-fuel damping uses the same cubic with the generated live-moisture ratio $r_l = M_l/M_{x,l}$:

$$\eta_{M,l} = \max\{0,\ 1 - 2.59r_l + 5.11r_l^2 - 3.52r_l^3\}.$$

For generated SH2, $M_{x,l} = 0.7080597$. At $M_l = 0.70$, $r_l = 0.70/0.7080597 = 0.988617$, giving $\eta_{M,l} = 0.0326506$; substitution in the weighted Rothermel calculation gives $I_R = 354.3991$ kW m^{-2} and $R = 0.0800849$ m min^{-1}. At $M_l = 0.90$, $r_l = 1.27108 > 1$, so the live contribution is zero and the remaining dead-fuel contribution gives $I_R = 304.8611$ kW m^{-2} and $R = 0.0579864$ m min^{-1}. The calculation is repeated for each generated SH2, SH6, and SH8 moisture value. Direct metrics use $e_R = |R_{\text{max}} - R_{\text{ref}}|/\max(|R_{\text{ref}}|, 10^{-8})$ and the analogous e_I, with $\max e_R \le 0.02$ and $\max e_I \le 0.01$. Where time of arrival (TOA) is required, $t = a + bx$ gives $R_{\text{TOA}} = 60/b$, $R^2 = 1 - SSE/SST$, $\max e_{\text{TOA}} \le 0.05$, and $\min R^2 \ge 0.995$.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS and 0.01 for reaction intensity. Simulation conditions predicted at or above 0.10 m min^{-1} also require TOA-derived ROS error at most 0.05 and minimum $R^2 = 0.995$; slower simulation conditions use the direct field because the practical run-duration cap cannot provide a multi-cell TOA regression. Pointwise analytical comparison is selected because a merely monotonic curve would not detect an incorrect live-extinction threshold. All 21 simulation conditions are required and the preprocessor rejects a fuel row marked dynamic.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE's supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive reference spread rate, the prescribed and maximum timesteps use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured Courant–Friedrichs–Lewy (CFL) limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE's stalled-front branch and corrupt the terminal timestamp. A zero-ROS simulation condition retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required simulation condition must reach the prescribed stop time in its generated namelist; a direct-only simulation condition may terminate early because lack of propagation is part of the state being tested. These shrub models carry live woody but no live herbaceous load. Live-woody moisture is set to 60, 70, 90, 120, 150, 180, or 200%; 60% is the supported ELMFIRE lower bound; dead classes remain at 8%. The homogeneous domain uses a finite ignition strip whose east face is planar in the interior measurement corridor and whose ends remain inside the lateral boundaries. Wind and slope are zero and all coupled processes are disabled. Fuel loads remain invariant; reaction intensity and ROS should generally decline as live moisture rises, with behavior governed by each fuel's calculated live extinction.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Fuel models	SH2, SH6, and SH8	code/name	Static live-fuel responses
Live moisture	30 to 200%	percent	Independent damping variable
Wind/slope	0 / 0	mph / degrees	Isolates live damping
Timing	generated whole-second step and exact-multiple stop	s	Terminal-evidence compatibility

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

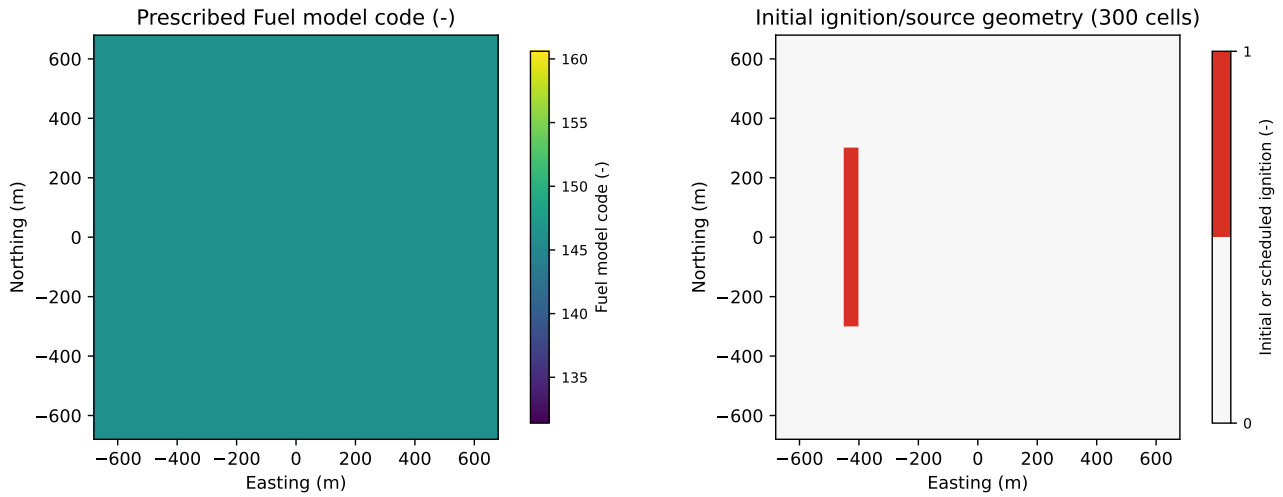


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The available results cover 21 of 21 required simulation conditions and reports **PASS**.

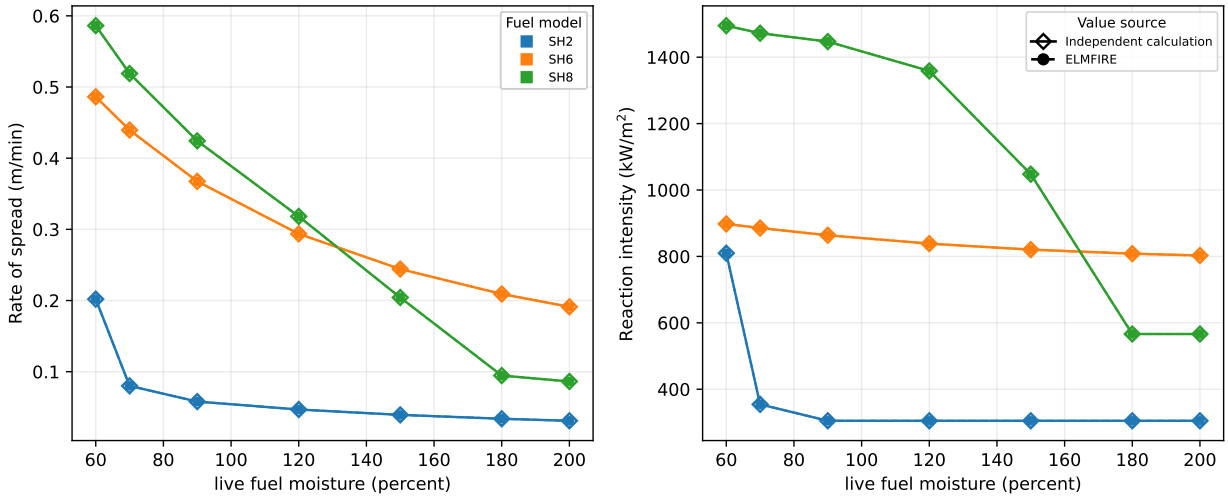


Figure 2: Independent calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines) for the static live-moisture response. Each fuel-model pair uses the same color.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	6.50329e-06 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	6.30732e-06 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.0170672 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999953 dimensionless	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

A shifted transition implicates calculated live extinction or moisture units. Unexpected sharp changes suggest that a supposedly static row was treated dynamically. Agreement in intensity but not ROS narrows the defect to heat sink or spread conversion.

9 Scope and limitations

This case covers three static shrub fuels at one dead-moisture state. It does not verify dynamic curing, spatial moisture mosaics, temporal interpolation, or physical observations. Dynamic herbaceous transfer is isolated in CASE39.